

Digital Design (SET 111)

L06

Boolean Algebra and Logic Simplification

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- ☐ **Boolean Operations and Expressions**
- ☐ **Laws of Boolean Algebra**
- ☐ **Rules of Boolean Algebra**
- ☐ **DeMorgan's Theorems**
- ☐ **Boolean Expression for a Logic Circuit**



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Boolean Operations and Expressions



□ Definition

□ Boolean algebra is the **mathematics** of digital logic

□ Variable and complement are terms widely used in Boolean algebra

□ A **variable** is a symbol used to represent an action, a condition, or data {0 ,1 }

□ The complement is the inverse of a variable and is indicated by a **bar** over the variable. The complement of the variable A is $\overline{A}$

□ Main Boolean Operation

□ Boolean **Addition** equivalent to the OR operation ($F=A+B$)

□ Boolean **Multiplication** is equivalent to the AND operation ($F=A.B$)



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Laws of Boolean Algebra (1/2)



☐ A) Commutative Law

- ☐ Addition for two variables $\rightarrow A+B=B+A$
- ☐ Multiplication for two variables $\rightarrow A.B=B.A$

☐ B) Associative Law

- ☐ Addition for multiple variables $\rightarrow A+(B+C)=(A+B)+C$
- ☐ Multiplication for multiple variables $\rightarrow A.(B.C)=(A.B).C$

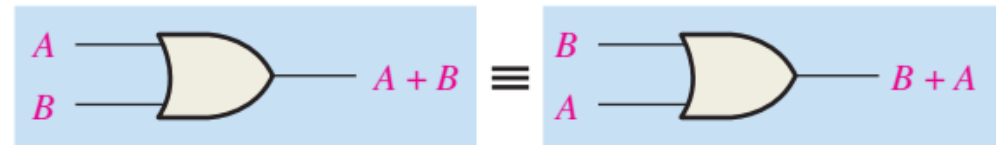
☐ C) Distributive Law

- ☐ The distributive law is written for three variables $\rightarrow A.(B+C)=A.B+A.C$

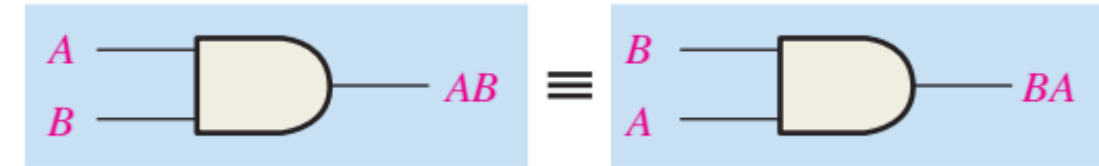
Laws of Boolean Algebra (2/2)



□ A) Commutative Law

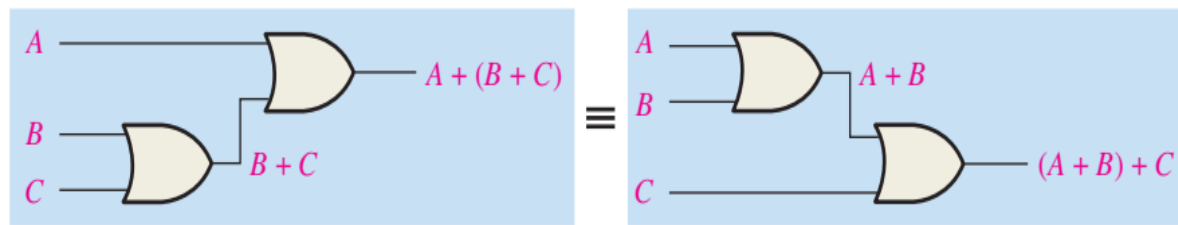


Application of commutative law of addition

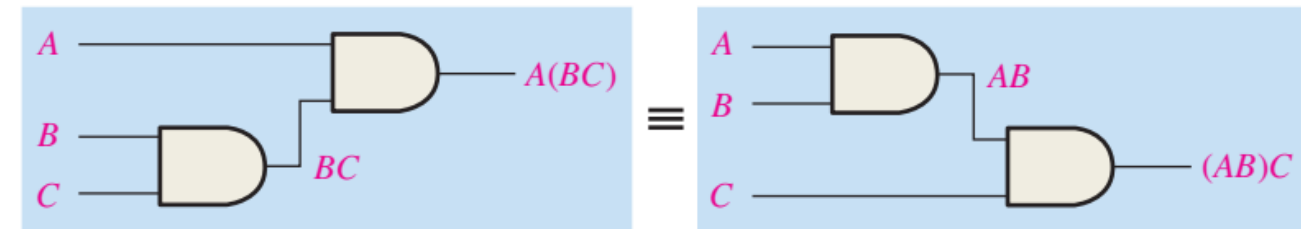


Application of commutative law of multiplication

□ B) Associative Law

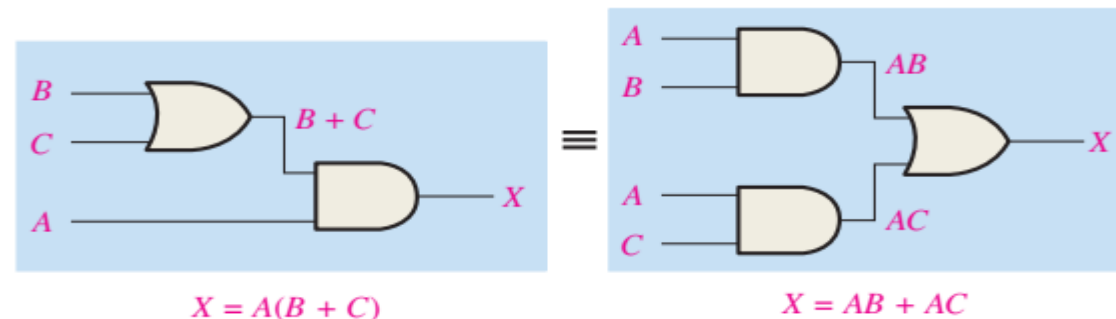


Application of associative law of addition



Application of associative law of multiplication

□ C) Distributive Law



$$X = A(B + C)$$

$$X = AB + AC$$

Application of distributive law



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Rules of Boolean Algebra



Rule 1: $A+0=A$

Rule 2: $A+1=1$

Rule 3: $A+A=A$

Rule 4: $A+\bar{A}=1$

Rule 5: $A.0=0$

Rule 6: $A.1=A$

Rule 7: $A.A=A$

Rule 8: $A.\bar{A}=0$

Rule 9: $\overline{\bar{A}}=A$



Rule 10: $A+A.B=A$

Rule 11: $(A+B).(A+C)=A+BC$

Rule 12: $(A+\bar{A}B)=A+B$



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DeMorgan's Theorems (1/2)



- ❑ **DeMorgan**, a mathematician who knew Boole proposed two theorems that are an important part of Boolean algebra
- ❑ **DeMorgan's** theorems provide mathematical verification of the equivalency of the NAND and NOR gates.
- ❑ **DeMorgan's first** theorem
 - ❑ The complement of a **product** of variables is equal to the sum of the complements of the variables.
- ❑ **DeMorgan's second** theorem
 - ❑ The complement of a **sum** of variables is equal to the product of the complements of the variables.

$$\overline{XY} = \overline{X} + \overline{Y}$$

$$\overline{X + Y} = \overline{X} \cdot \overline{Y}$$

DeMorgan's Theorems (2/2)



□ Ex.1. Apply DeMorgan's theorems to the expressions $\overline{X.Y.Z}$

Ans

$$\overline{X.Y.Z} = \overline{X} + \overline{Y} + \overline{Z}$$

□ Ex.2. Apply DeMorgan's theorems to the expressions $\overline{X + Y + Z}$

Ans

$$\overline{X + Y + Z} = \overline{X} . \overline{Y} . \overline{Z}$$

□ Ex.3. Apply DeMorgan's theorems to the expressions $\overline{A . \overline{B} + \overline{C} . D + E . F}$

Ans

$$\overline{A . \overline{B} + \overline{C} . D + E . F} = (\overline{A} + B) . (C + \overline{D}) . (\overline{E} + \overline{F})$$

□ Ex.4. Apply DeMorgan's theorems to the expressions $\overline{(A + B + C) . D}$

Ans

$$\overline{(A + B + C) . D} = \overline{A} . \overline{B} . \overline{C} + \overline{D}$$



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Boolean Expression for a Logic Circuit (1/4)

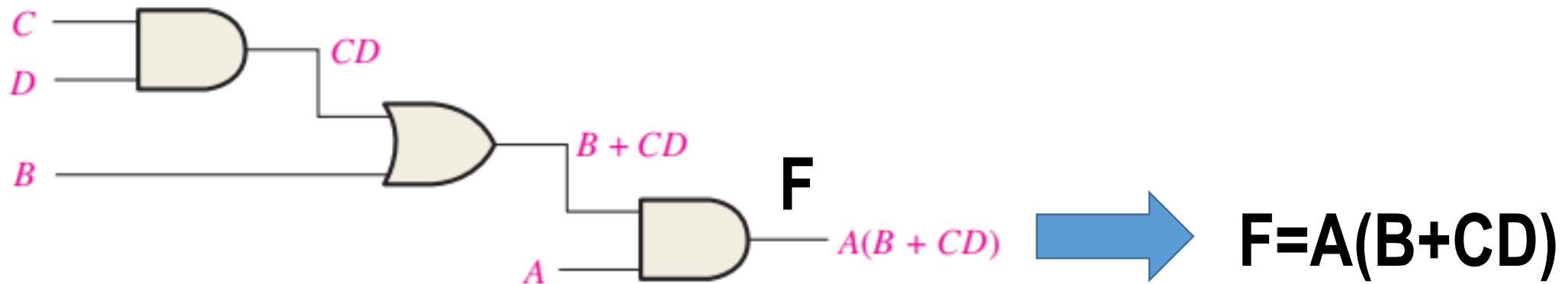


□ Main Steps

□ **Begin** at the left-most inputs

□ **Move** on toward the final output meanwhile writing the expression for each gate.

□ **EX.1.** Determine the Boolean expression for the following logic circuit.



Boolean Expression for a Logic Circuit (2/4)



□ Ex.1 Using Boolean algebra techniques, simplify this expression

$$AB + A(B + C) + B(B + C)$$

Ans

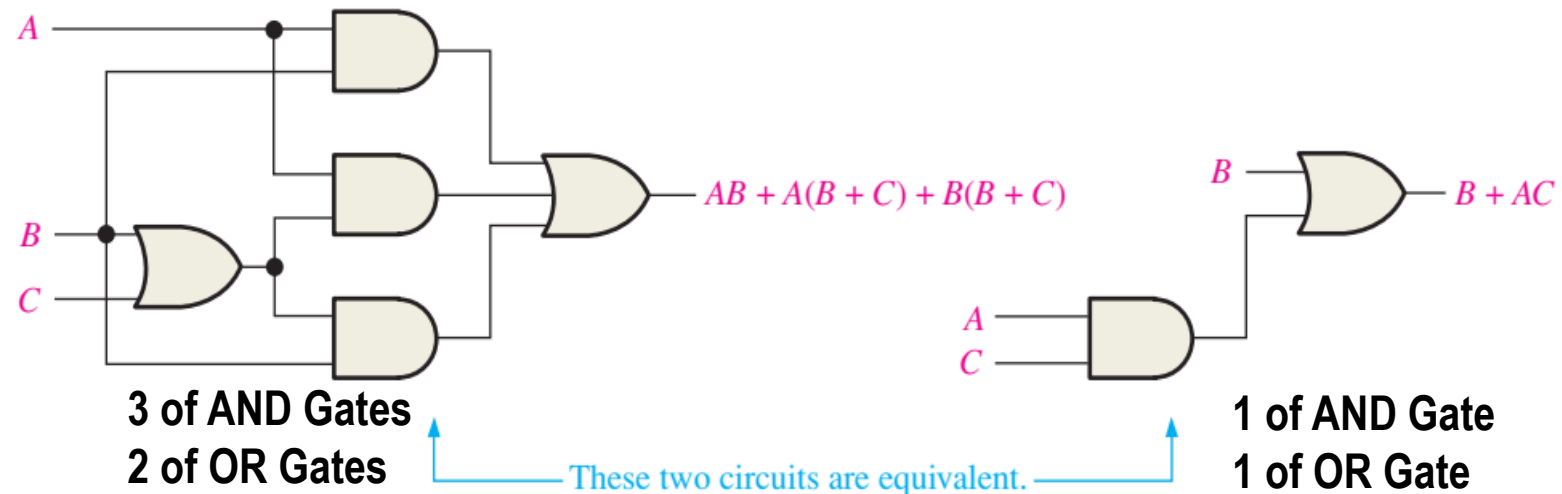
$$AB + A(B + C) + B(B + C) = \mathbf{AB + AB + AC + BB + BC} \quad \longrightarrow \quad \text{Where } AB + AB = AB$$

$$= AB + AC + \mathbf{BB} + BC \quad \longrightarrow \quad \text{Where } BB = B$$

$$= AB + AC + \mathbf{B + BC} \quad \longrightarrow \quad \text{Where } B + BC = B(1 + C) = B$$

$$= \mathbf{AB + AC + B} \quad \longrightarrow \quad \text{Where } AB + B = B(1 + A) = B$$

$$= \mathbf{AC + B}$$



Boolean Expression for a Logic Circuit (3/4)



□ Ex.2 Using Boolean algebra techniques, simplify this expression

$$[A \overline{B} (C + BD) + \overline{A} \overline{B}]C$$

Ans

$$[A \overline{B} (C + BD) + \overline{A} \overline{B}]C = [A \overline{B} C + A \overline{B} BD + \overline{A} \overline{B}]C$$

$$= A \overline{B} CC + A \overline{B} BDC + \overline{A} \overline{B} C \quad \longrightarrow \text{Where } C.C=C, \overline{B}B=0$$

$$= A \overline{B} C + 0 + \overline{A} \overline{B} C \quad \longrightarrow \text{Where } C.C=C, \overline{B}B=0$$

$$= \overline{B} C (A + \overline{A}) \quad \longrightarrow \text{Where } (A + \overline{A})=1$$

$$= \overline{B} C$$

Boolean Expression for a Logic Circuit (4/4)



□ Ex.3 Using Boolean algebra techniques, simplify this expression

$$\overline{AB + AC} + \overline{A} \overline{B} C$$

Ans

$$\begin{aligned}\overline{AB + AC} + \overline{A} \overline{B} C &= (\overline{A} + \overline{B}).(\overline{A} + \overline{C}) + \overline{A} \overline{B} C \\&= (\overline{A} \overline{A} + \overline{A} \overline{C} + \overline{A} \overline{B} + \overline{B} \overline{C}) + \overline{A} \overline{B} C \quad \longrightarrow \text{Where } \overline{A} \overline{A} = \overline{A} \\&= \overline{A} + \overline{A} \overline{C} + \overline{A} \overline{B} + \overline{B} \overline{C} + \overline{A} \overline{B} C \quad \longrightarrow \text{Where } \overline{A} (1 + \overline{C}) = \overline{A} \\&= \overline{A} + \overline{A} \overline{B} + \overline{B} \overline{C} + \overline{A} \overline{B} C \quad \longrightarrow \text{Where } \overline{A} \overline{B} (1 + C) = \overline{A} \overline{B} \\&= \overline{A} + \overline{B} \overline{C} + \overline{A} \overline{B} \quad \longrightarrow \text{Where } \overline{A} (1 + \overline{B}) = \overline{A} \\&= \overline{A} + \overline{B} \overline{C}\end{aligned}$$

Thanks!